Generative Modelling with Stochastic Differential Equations

Exercises, week 3

Time reversal

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4 Noising and time reversal

Setting and facts used in these exercises. $\gamma_n = \mathcal{N}(0, \mathbb{I}_n)$ is the standard Gaussian on $\mathbb{R}^n$. The *Ornstein–Uhlenbeck (OU) process* is the solution of

$$\mathrm{d}\bar{X}_\tau = -\bar{X}_\tau \,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau, \qquad \tau \geq 0,$$

with $\bar{X}_0 \sim q$ independent of the Brownian motion B. With $\sigma_\tau^2 := 1 - e^{-2\tau}$, its solution is $\bar{X}_\tau = e^{-\tau}\bar{X}_0 + \sqrt{2}\int_0^\tau e^{-(\tau-r)}\,\mathrm{d}B_r$, so that $\mathrm{Law}(\bar{X}_\tau \mid \bar{X}_0 = x) = \mathcal{N}(e^{-\tau}x, \sigma_\tau^2 \mathbb{I}_n)$ and $\bar{X}_\tau \overset{d}{=} e^{-\tau}\bar{X}_0 + \sigma_\tau Z$ with $Z \sim \gamma_n$ independent of $\bar{X}_0$. We write $q_\tau = \mathrm{Law}(\bar{X}_\tau)$ (a density for $\tau > 0$) and $s_\tau = \nabla \log q_\tau$ for its *score*. The OU process has invariant law γ_n, and for every initial law

$$\chi^2(q_\tau \,\|\, \gamma_n) \leq e^{-2\tau}\chi^2(q \,\|\, \gamma_n), \qquad \chi^2(\mu \,\|\, \nu) = \int \Big(\frac{\mathrm{d}\mu}{\mathrm{d}\nu} - 1\Big)^2 \mathrm{d}\nu \;\; (= \infty \text{ if } \mu \not\ll \nu). \tag{$*$}$$

For a horizon $T > 0$, the *reverse OU process* is the solution on $[0, T)$ of

$$\mathrm{d}\overleftarrow{X}_t = \big[\overleftarrow{X}_t + 2 s_{T-t}(\overleftarrow{X}_t)\big]\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t, \tag{$**$}$$

and the time-reversal theorem states that $(\bar{X}_{T-t})_{t\in[0,T)}$ is a solution of $(**)$ started from q_T (for a suitable Brownian motion $\tilde{B}$). $\mathrm{TV}(\mu,\nu) = \sup_A |\mu(A) - \nu(A)|$ is the total variation distance.

Exercise 4.1 (Time changes in simulated annealing)**.** Let $U : \mathbb{R}^n \to \mathbb{R}$ be smooth with ∇U globally Lipschitz, and let $\beta : [0,\infty) \to (0,\infty)$ be continuous with $\inf \beta > 0$. Let X solve

$$\mathrm{d}X_t = -\beta(t)\nabla U(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t.$$

Let $S(t) = \int_0^t \beta(r)\,\mathrm{d}r$ and let $t(s)$ be the inverse function of S.

(a) Show that $W_s := \int_0^{t(s)} \sqrt{\beta(r)}\,\mathrm{d}B_r$ is a Brownian motion with respect to $(\mathcal{F}_{t(s)})_{s\geq 0}$.

(b) Show that $Y_s := X_{t(s)}$ solves $\mathrm{d}Y_s = -\nabla U(Y_s)\,\mathrm{d}s + \sqrt{2/\tilde{\beta}(s)}\,\mathrm{d}W_s$ with $\tilde{\beta}(s) = \beta(t(s))$. In particular, for constant β, $Y_s = X_{s/\beta}$, so putting the inverse temperature on the drift, $\mathrm{d}X_t = -\beta\nabla U(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, or on the noise, $\mathrm{d}Y_s = -\nabla U(Y_s)\,\mathrm{d}s + \sqrt{2/\beta}\,\mathrm{d}W_s$, are equivalent.

(c) For constant β, check that $\pi_\beta \propto e^{-\beta U}$ (assumed normalizable) is invariant for both SDEs.

Hint for (a): by Lévy's characterisation, a continuous martingale W with $W_0 = 0$ and quadratic variation $\langle W\rangle_s = s\,\mathbb{I}_n$ is a Brownian motion. Since $t(\cdot)$ is deterministic, evaluating a martingale at the times $t(s)$ gives a martingale in the filtration $(\mathcal{F}_{t(s)})_s$.

Exercise 4.2 (OU process with a general rate)**.** Let $\theta > 0$ and let X solve $\mathrm{d}X_\tau = -\theta X_\tau\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau$ on $\mathbb{R}^n$, with X_0 independent of B.

(a) Show that $X_\tau = e^{-\theta\tau}X_0 + \sqrt{2}\int_0^\tau e^{-\theta(\tau-r)}\,\mathrm{d}B_r$.

(b) Show that $\mathrm{Law}(X_\tau \mid X_0 = x) = \mathcal{N}(e^{-\theta\tau}x,\ \theta^{-1}(1 - e^{-2\theta\tau})\mathbb{I}_n)$, and that $\mathcal{N}(0, \theta^{-1}\mathbb{I}_n)$ is invariant. Which potential U makes this a Langevin diffusion?

(c) Show that $Y_\tau := \sqrt{\theta}\, X_{\tau/\theta}$ is an OU process with rate 1, i.e. solves $\mathrm{d}Y_\tau = -Y_\tau\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}W_\tau$ for a Brownian motion W.

(d) Deduce that $\chi^2(\,\mathrm{Law}(X_\tau)\,\|\,\mathcal{N}(0,\theta^{-1}\mathbb{I}_n)) \le e^{-2\theta\tau}\chi^2(\,\mathrm{Law}(X_0)\,\|\,\mathcal{N}(0,\theta^{-1}\mathbb{I}_n))$, and compare with the rate $2/C_{\mathrm{P}}$ predicted by the Poincaré constant of the invariant measure.

Exercise 4.3 (Exact χ^2 for a Gaussian start)**.** Let $m \in \mathbb{R}^n$ and $v > 0$.

(a) Show that

$$\chi^2\big(\mathcal{N}(m, v\mathbb{I}_n)\,\|\,\gamma_n\big) = \begin{cases} \big(v(2-v)\big)^{-n/2}\exp\big(|m|^2/(2-v)\big) - 1, & v < 2, \\ +\infty, & v \ge 2. \end{cases}$$

(b) Let $q = \mathcal{N}(m, v\mathbb{I}_n)$ and let q_τ be the OU marginals. Compute $\chi^2(q_\tau\,\|\,\gamma_n)$ exactly.

(c) Take $v = 1$. Check $(*)$ directly, and show that $\chi^2(q_\tau\,\|\,\gamma_n)/\chi^2(q\,\|\,\gamma_n) \to e^{-2\tau}$ as $m \to 0$, and deduce that the rate 2 is sharp.

(d) Take $m = 0$, $v \neq 1$. Show that $\chi^2(q_\tau\,\|\,\gamma_n)$ decays like $e^{-4\tau}$. If $v \ge 2$, show that $\chi^2(q\,\|\,\gamma_n) = \infty$ while $\chi^2(q_\tau\,\|\,\gamma_n) < \infty$ for $\tau > \frac{1}{2}\log(v-1)$.

Exercise 4.4 (Tweedie's formula and the score of a Gaussian mixture)**.** Let $\bar{X}_0 \sim q$ be an arbitrary probability measure on $\mathbb{R}^n$, $Z \sim \gamma_n$ independent, $\tau > 0$, and $\bar{X}_\tau = e^{-\tau}\bar{X}_0 + \sigma_\tau Z$, whose density is $q_\tau(x) = \int (2\pi\sigma_\tau^2)^{-n/2}\exp\big(-|x - e^{-\tau}z|^2/(2\sigma_\tau^2)\big)\,q(\mathrm{d}z)$.

(a) Deduce from Tweedie's formula of Chapter 2 (Proposition 2.15) that

$$s_\tau(x) = \nabla \log q_\tau(x) = -\frac{1}{\sigma_\tau^2}\big(x - e^{-\tau}\mathbb{E}[\bar{X}_0 \mid \bar{X}_\tau = x]\big).$$

(b) Prove the noise form $s_\tau(x) = -\sigma_\tau^{-1}\mathbb{E}[Z \mid \bar{X}_\tau = x]$, and deduce that s_τ minimizes $f \mapsto \mathbb{E}|f(\bar{X}_\tau) + Z/\sigma_\tau|^2$ over measurable f with $\mathbb{E}|f(\bar{X}_\tau)|^2 < \infty$.

(c) Let $q = \sum_{k=1}^K w_k\mathcal{N}(m_k, C_k)$ with $w_k > 0$, $\sum_k w_k = 1$ and C_k positive semidefinite. Show that $q_\tau = \sum_k w_k\mathcal{N}(e^{-\tau}m_k, C_k(\tau))$ with $C_k(\tau) = e^{-2\tau}C_k + \sigma_\tau^2\mathbb{I}_n$, and

$$s_\tau(x) = -\sum_{k=1}^K \pi_k(\tau,x)\,C_k(\tau)^{-1}(x - e^{-\tau}m_k), \qquad \pi_k(\tau,x) = \frac{w_k\mathcal{N}(x; e^{-\tau}m_k, C_k(\tau))}{q_\tau(x)}.$$

Check that this agrees with Tweedie's formula.

(d) Let e be a unit vector and $q = w_-\mathcal{N}(-me, v\mathbb{I}_n) + w_+\mathcal{N}(me, v\mathbb{I}_n)$. Only the coordinate $\langle e, x\rangle$, in the direction separating the modes, enters the nonlinear part. Show that

$$s_\tau(x) = -\frac{x}{v_\tau} + \frac{e^{-\tau}m}{v_\tau}\tanh\Big(\frac{e^{-\tau}m\langle e, x\rangle}{v_\tau} + \frac{1}{2}\log\frac{w_+}{w_-}\Big)e, \qquad v_\tau = 1 + (v-1)e^{-2\tau}.$$

Exercise 4.5 (Gaussian reversal from the wrong initial law)**.** Let $q = \mathcal{N}(m, v\mathbb{I}_n)$ with $v > 0$, and $v_\tau = 1 + (v-1)e^{-2\tau}$. By Example 4.2 with $S = v\mathbb{I}_n$, the reverse SDE $(**)$ is the linear SDE

$$\mathrm{d}\overleftarrow{X}_t = \Big[\overleftarrow{X}_t - \frac{2}{v_{T-t}}\big(\overleftarrow{X}_t - e^{-(T-t)}m\big)\Big]\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t,$$

and its solution started from q_T has law q_{T-t} at time t.

(a) Start the same SDE from $\hat{Y}_0 \sim \gamma_n$, independent of $\tilde{B}$. Show that

$$\mathrm{Law}(\hat{Y}_T) = \mathcal{N}\Big(m - \frac{mv}{e^{2T}+v-1},\ \Big(v - \frac{(v-1)v^2}{(e^{2T}+v-1)^2}\Big)\mathbb{I}_n\Big).$$

(b) For fixed n and $m \neq 0$, compare the size of the error with the bound $\mathrm{TV}(\mathrm{Law}(\hat{Y}_T), q) \leq \mathrm{TV}(\gamma_n, q_T)$. It follows from the proof of Proposition 4.14 with $\delta = 0$, since for $v > 0$ the drift is Lipschitz on $[0, T]$.

(c) What happens as $v \to 0$?

Hint: $2/v_\tau = \frac{\mathrm{d}}{\mathrm{d}\tau} \log(e^{2\tau} + v - 1)$.

Exercise 4.6 (DDPM and OU time)**.** Let $\beta_1, \dots, \beta_K \in (0,1)$, $x_0 \sim q$, and $x_k = \sqrt{1-\beta_k}\, x_{k-1} + \sqrt{\beta_k}\, \xi_k$ for $k \geq 1$, with $\xi_k \overset{\text{i.i.d.}}{\sim} \gamma_n$ independent of x_0. Let $\bar{\alpha}_k = \prod_{i \leq k}(1-\beta_i)$ and $\tau_k = -\frac{1}{2}\log \bar{\alpha}_k$.

(a) Show that $x_k \overset{d}{=} \sqrt{\bar{\alpha}_k}\, x_0 + \sqrt{1-\bar{\alpha}_k}\, Z$ with $Z \sim \gamma_n$ independent of x_0.

(b) Show that $(x_0, \dots, x_K) \overset{d}{=} (\bar{X}_{\tau_0}, \dots, \bar{X}_{\tau_K})$, where $\bar{X}$ is the OU process with $\bar{X}_0 \sim q$. In particular $e^{-2\tau_k} = \bar{\alpha}_k$ and $\sigma^2_{\tau_k} = 1 - \bar{\alpha}_k$.

(c) Let $\beta : [0, S] \to (0, \infty)$ be continuous. Show that the VP-SDE $\mathrm{d}X_s = -\frac{1}{2}\beta(s) X_s\, \mathrm{d}s + \sqrt{\beta(s)}\, \mathrm{d}W_s$, $X_0 \sim q$, satisfies $(X_s)_s \overset{d}{=} (\bar{X}_{\tau(s)})_s$ with $\tau(s) = \frac{1}{2}\int_0^s \beta(r)\, \mathrm{d}r$. Show that if $\beta_k = \beta(s_k)(s_k - s_{k-1})$ on a grid $0 = s_0 < \cdots < s_K = S$ of mesh η, then $\max_k |\tau_k - \tau(s_k)| \to 0$ as $\eta \to 0$.

(d) Show that the minimizer of $\mathbb{E}|\hat{\varepsilon}(\sqrt{\bar{\alpha}_k} x_0 + \sqrt{1-\bar{\alpha}_k} Z) - Z|^2$ over measurable $\hat{\varepsilon}$ is $\hat{\varepsilon} = -\sigma_{\tau_k} s_{\tau_k}$.

Exercise 4.7 (Initialisation with a finite second moment)**.** Let $T \geq 1$, $\delta \in (0, T)$ and $m_2^2 = \int |x|^2\, q(\mathrm{d}x) < \infty$. Let $\hat{Y}$ solve $(**)$ with the exact score and $\hat{Y}_0 \sim \gamma_n$. Assume the SDE is well posed on $[0, T-\delta]$ from every initial law.

(a) Show that a Markov kernel K contracts total variation: $\mathrm{TV}(\mu K, \nu K) \leq \mathrm{TV}(\mu, \nu)$. Explain why $\mathrm{Law}(\hat{Y}_{T-\delta}) = \gamma_n K$ and $q_\delta = q_T K$ for the same kernel.

(b) Use Pinsker's inequality and Lemma 4.13 to show that

$$\mathrm{TV}\big(\mathrm{Law}(\hat{Y}_{T-\delta}), q_\delta\big) \leq e^{-T}\sqrt{\tfrac{1}{4}m_2^2 + \tfrac{n}{2}e^{-2T}}.$$

Deduce the bound $\sqrt{n/2}\, e^{-T}$ when $m_2^2 \leq n$.

(c) Apply the first bound in (b) to $q = \delta_m$. Why can it be used even though q has no density?

General reversal. For the remaining exercises, let $\mathrm{d}X_\tau = b_\tau(X_\tau)\, \mathrm{d}\tau + g_\tau\, \mathrm{d}B_\tau$, where g_τ is independent of position. Write $\Sigma_\tau = g_\tau g_\tau^\top$, $q_\tau = \mathrm{Law}(X_\tau)$ and $\overleftarrow{X}_t = X_{T-t}$. Under Assumption 4.3, the reverse equation on $[0, T)$ is

$$\mathrm{d}\overleftarrow{X}_t = \overleftarrow{b}_t(\overleftarrow{X}_t)\, \mathrm{d}t + g_{T-t}\, \mathrm{d}\tilde{B}_t, \qquad \overleftarrow{b}_t = -b_{T-t} + \Sigma_{T-t}\nabla \log q_{T-t}. \tag{TR}$$

The Brownian motion $\tilde{B}$ is adapted to the reverse filtration, possibly enlarged.

Exercise 4.8 (Transition densities of the reversal)**.** Let $(X_\tau)_{\tau\in[0,T]}$ be a Markov process on $\mathbb{R}^n$ with positive continuous transition densities $p_{r,\tau}(x,y)$, $0\le r<\tau\le T$, and positive marginal densities q_τ, $\tau\in(0,T]$. Let $\overleftarrow{X}_t = X_{T-t}$ and, for $0\le s<t<T$,

$$\overleftarrow{p}_{s,t}(x,y) = \frac{p_{T-t,T-s}(y,x)\,q_{T-t}(y)}{q_{T-s}(x)}.$$

(a) Show directly that $\int \overleftarrow{p}_{s,t}(x,y)\,\mathrm{d}y = 1$ and that the Chapman–Kolmogorov equation $\int \overleftarrow{p}_{s,t}(x,y)\,\overleftarrow{p}_{t,u}(y,z)\,\mathrm{d}y = \overleftarrow{p}_{s,u}(x,z)$ holds for $s<t<u<T$.

(b) Show that for $0\le t_0<\cdots<t_k<T$, with $\tau_j = T-t_j$, the joint density of $(\overleftarrow{X}_{t_0},\dots,\overleftarrow{X}_{t_k})$ at $(x_0,\dots,x_k)$ is

$$q_{\tau_k}(x_k)\prod_{j=1}^{k} p_{\tau_j,\tau_{j-1}}(x_j,x_{j-1}) = q_{T-t_0}(x_0)\prod_{j=1}^{k}\overleftarrow{p}_{t_{j-1},t_j}(x_{j-1},x_j).$$

Deduce, without using any abstract characterisation of the Markov property, that $\overleftarrow{X}$ is a Markov process on $[0,T)$ with transition densities $\overleftarrow{p}$.

(c) Show by an example that the reversal of a time-homogeneous Markov process need not be time-homogeneous.

Exercise 4.9 (Reverse drift via the weighted adjoint)**.** Let $\mathcal{L}_\tau\varphi = \langle b_\tau,\nabla\varphi\rangle + \frac12\operatorname{tr}(\Sigma_\tau\nabla^2\varphi)$, where Σ_τ is symmetric and independent of x. Assume b_τ is C^1 and $q_\tau>0$ is C^2.

(a) Write down the Lebesgue adjoint $\mathcal{L}_\tau^*$.

(b) Expand $q_\tau^{-1}[\mathcal{L}_\tau^*(q_\tau\psi) - \psi\mathcal{L}_\tau^* q_\tau]$. Show that it has diffusion matrix Σ_τ and drift $-b_\tau + \Sigma_\tau\nabla\log q_\tau$.

(c) Evaluate the reverse drift for OU.

(d) Check that reversing twice returns the original drift.

Exercise 4.10 (The Brownian bridge as the reversal of Brownian motion)**.** Let $x_0\in\mathbb{R}^n$, $X_\tau = x_0 + B_\tau$ for $\tau\in[0,T]$, and $\overleftarrow{X}_t = X_{T-t}$.

(a) Use (TR) to show that $\mathrm{d}\overleftarrow{X}_t = -\frac{\overleftarrow{X}_t - x_0}{T-t}\,\mathrm{d}t + \mathrm{d}\tilde{B}_t$ on $[0,T)$, with $\overleftarrow{X}_0\sim\mathcal{N}(x_0,T\mathbb{I}_n)$.

(b) Solve this SDE explicitly. Show that, conditionally on $\overleftarrow{X}_0 = y$, $\overleftarrow{X}$ is a Gaussian process with mean $x_0 + \frac{T-t}{T}(y-x_0)$ and covariance $\operatorname{Cov}(\overleftarrow{X}_s,\overleftarrow{X}_t) = \frac{s(T-t)}{T}\mathbb{I}_n$ for $s\le t$, i.e. a Brownian bridge from y to x_0. Check this against a direct Gaussian conditioning computation for (X_{T-s},X_{T-t}) given $X_T = y$.

(c) Let $W_t := B_T - B_{T-t}$. Show that W is a Brownian motion and that $\overleftarrow{X}_t = \overleftarrow{X}_0 - W_t$, a representation without drift. Why does this not contradict (a)? Express $\tilde{B}$ in terms of W.

Hint for (c): a Brownian motion in the filtration of $\overleftarrow{X}$ would be independent of $\overleftarrow{X}_0$. Compute $\operatorname{Cov}(\overleftarrow{X}_0, W_t)$.

Exercise 4.11 (Reversal of stationary diffusions)**.**

(a) Let $\mu\propto e^{-U}$ on $\mathbb{R}^n$ and let X solve $\mathrm{d}X_\tau = -\nabla U(X_\tau)\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau$ with $X_0\sim\mu$. Assuming (TR) applies and the SDE is unique in law, show that $(X_{T-t})_{t\in[0,T]}$ and $(X_t)_{t\in[0,T]}$ have the same law.

(b) Let $A \neq 0$ be an antisymmetric $n \times n$ matrix and let X solve $\mathrm{d}X_\tau = -(\mathbb{I}_n + A)X_\tau \,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau$. Show that γ_n is invariant.

(c) With $X_0 \sim \gamma_n$, compute the drift of the reversed process and show that it differs from the forward drift. Interpret this in terms of the generator in $L^2(\gamma_n)$.